



FIRST SEMESTER, 2026-2027

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : BITS F446
Course Title : Pattern Recognition
Instructor-in-Charge : Raj Kumar Gupta
Instructors : NA

1. Course Description:

This is a first course on Pattern recognition. It begins with an introduction to the fundamentals of digital images and develops the theoretical background for the pattern recognition. It goes on to discuss the different aspects like the preprocessing of a digital image, segmentation, representation and description and finally the object recognition. During the course, a flavour of matlab programming will also be given.

2. Scope & Objective of the Course:

Over the past 30 years or so, pattern recognition and automation have evolved into a mature subject embracing many topics and applications: these range from automatic (robot) assembly to automatic vehicle guidance, from automatic interpretation of documents to verification of signatures, and from analysis of remotely sensed images to checking of fingerprints and human blood cells. Currently, pattern recognition and automatic visual inspection are undergoing very substantial growth, necessary improvement in quality, safety and cost effectiveness being the stimulating factors. The course introduces the students to the fundamentals of digital images and various processing techniques that are applied to them not only to improve their quality but also to recognize them.

3. Textbooks:

TB1. *Digital Image Processing*, R. C. Gonzalez and R. E. Woods, 3rd and above, Pearson Education Asia, 2007.

References:

Ref1. *Image Processing, analysis and machine vision*, M. Sonka, V. Hlavac and R. Boyle, 2nd Ed., Thompson Asia, 2003.

4. Course Plan:

Weekly plan for 14 weeks with [3] lectures per week

Module No.	Lecture Session	Reference	Learning Outcomes
Introduction to field L:1	Introduction to pattern recognition and application	Lecture notes	A general flow of the course structure complemented with examples
Foundation for digital image L:2–3	Digital image, data structure for image analysis	Lecture notes	Interpretation of data relevant to digital images
Preprocessing L:4–6	Preprocessing of the digital image	Ch. 3 & 4	Some basic gray level transformations for image enhancement and noise reduction
Introduction to software L:7–10	Introduction to Matlab	Lecture notes	Matlab programming relevant to image analysis for pattern recognition
Segmentation L:11–30	Segmentation I (L:11–19)	Ch. 10	Detection of discontinuities, edge-linked boundary detection, thresholding, region-based segmentation, segmentation by morphological watersheds, use of motion in segmentation
	Segmentation II (L:20–26)	Ch. 9	Morphological operations to segment out objects of interest
	Representation & Description (L:27–30)	Ch. 11	Representation of data and defining descriptors for a given problem
Classification & Training L:31–40	Object recognition	Ch. 12	Different classifiers (e.g., MDC, statistical, neural network, etc.). Examples of pattern recognition algorithms for a variety of systems

Table 1: Course Plan: Pattern Recognition and Image Analysis

5. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Surprise Quizzes	6	2 mins.	During Lecture	Closed Book, [†] best n-1 of n
Project	2	Take home	TBA	Open Book / Submission
Assignment	2	Take home.	TBA	Open Book
Tutorial Tests	20	20 mins	Lecture class	Closed Book, [†] best n-1 of n
Mid-Semester Test	30	90 mins.	TBA	Closed Book
Comprehensive Examination	40	180 mins.	TBA	Closed/Open Book

[†] n represents total number of quizzes/Tut tests taken.

Tentative Schedule for Tutorial Tests:

- Tut Test #1: Second last week of August
- Tut Test #2: First week of September
- Tut Test #3: Third week of September
- Tut Test #4: Last week of September
- Tut Test #5: Last week of October
- Tut Test #6: First week of November
- Tut Test #7: Third week of November

NC Criterion: In addition to the Institute's rules, marks obtained < 10% of the total weightage of the course.

6. Office Consultation Hour:

Will be announced in the class. Also, feel free to email me to seek a prior appointment.

7. Notice:

google classroom/nalanda (<https://nalanda-aws.bits-pilani.ac.in/>)

8. Make-up Policy:

Make-up tests will be given only to genuine cases. No make-up will be given for the tutorial tests.

signature

Raj Kumar Gupta

Instructor-in-charge

**BITS F446 and Pattern
Recognition**